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Energy, Environment and Microbiology

Edexcel A A-Level Biology — Topic 4

4.1 Photosynthesis

Photosynthesis: The process by which light energy is used to convert carbon dioxide and water into glucose and oxygen. It occurs in chloroplasts of plant cells and algae. It is a two-stage process: the light-dependent reactions (thylakoid membranes) produce ATP and reduced NADP, which are then used in the light-independent reactions (stroma) to fix carbon dioxide into organic molecules.

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2 \text{ (light energy required)}$$

Chloroplast Structure

Chloroplasts have a double outer membrane (envelope) that controls what enters and leaves. Inside is an internal system of **thylakoid membranes** arranged in stacks called **grana** (singular: granum), connected by **intergranal lamellae** (stroma lamellae). The fluid-filled **stroma** contains enzymes (including RuBisCO), circular DNA, 70S ribosomes, starch grains, and lipid droplets.

Key structural-functional relationships:

- **Thylakoid membranes** — provide a large surface area for photosynthetic pigments, electron carriers, and ATP synthase; the enclosed thylakoid lumen (space) allows a proton gradient to form for chemiosmosis
- **Grana** — stacking increases the membrane area for light-dependent reactions in a compact space
- **Stroma** — contains all enzymes needed for the Calvin cycle (light-independent reactions)

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- **Own DNA and ribosomes** — can synthesise some of their own proteins; evidence for the endosymbiotic theory (chloroplasts evolved from photosynthetic cyanobacteria engulfed by ancestral eukaryotic cells)

Photosynthetic Pigments

Plants contain several pigments that absorb different wavelengths of light, broadening the range of light that can drive photosynthesis:

Pigment	Colour	Wavelengths Absorbed	Role
Chlorophyll a	Blue-green	Red (~680 nm) and blue (~440 nm)	Primary pigment; found in reaction centres of PSI and PSII; directly involved in light-dependent reactions
Chlorophyll b	Yellow-green	Red (~650 nm) and blue (~470 nm)	Accessory pigment; absorbs wavelengths chlorophyll a absorbs poorly; passes energy to chlorophyll a
Carotenoids (carotene, xanthophyll)	Orange/yellow	Blue-violet (~400–500 nm)	Accessory pigments; broaden absorption spectrum; also protect chlorophyll from photooxidation

Pigments can be separated by **chromatography** (thin-layer or paper). The R_f value = distance moved by pigment / distance moved by solvent front. Carotene has the highest R_f (most soluble in the solvent); chlorophyll b has the lowest.

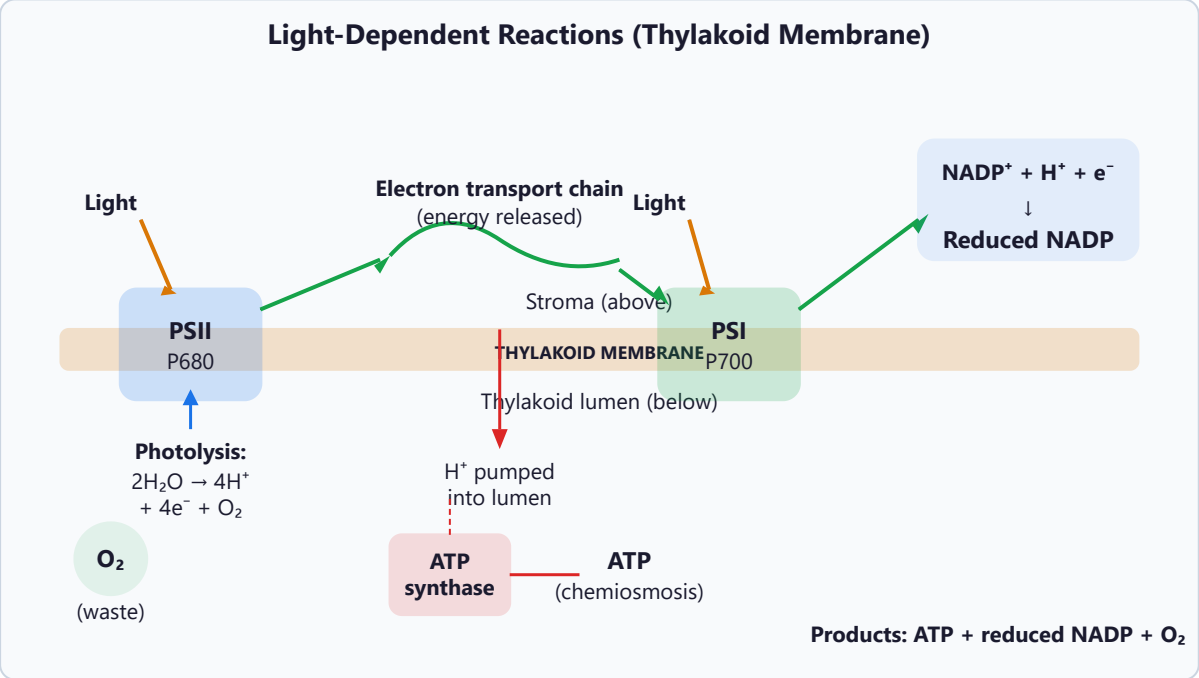
Light-Dependent Reactions (Thylakoid Membranes)

1. **Photoexcitation:** Light energy is absorbed by chlorophyll a in photosystem II (PSII, P680 — named after its absorption peak at 680 nm). Electrons in the chlorophyll are excited to a higher energy level and are captured by an electron acceptor — the chlorophyll is now oxidised
2. **Photolysis of water:** To replace the lost electrons, water is split by the oxygen-evolving complex associated with PSII: $2H_2O \rightarrow 4H^+ + 4e^- + O_2$. This is the source of all the oxygen released in

photosynthesis and occurs in the thylakoid lumen.

3. **Electron transport chain:** Excited electrons pass along a chain of electron carriers (including plastoquinone, cytochrome b6f complex, and plastocyanin), releasing energy at each step. This energy is used to pump H^+ ions from the stroma into the thylakoid lumen, building up a **proton gradient** (chemiosmotic gradient).
4. **ATP synthesis (photophosphorylation):** H^+ ions flow back through **ATP synthase** enzymes embedded in the thylakoid membrane (down the concentration gradient, from lumen to stroma). This drives the rotary mechanism of ATP synthase, catalysing ADP + P_i → ATP.
5. **Photosystem I (PSI, P700):** Light energy re-excites electrons arriving from the ETC via plastocyanin. These high-energy electrons are passed to ferredoxin.
6. **NADP reduction:** The enzyme **$NADP^+$ reductase** transfers the electrons from ferredoxin, along with H^+ ions from the stroma, to $NADP^+$, forming **reduced NADP (NADPH)**.

Products of the light-dependent reactions: ATP, reduced NADP (NADPH), and O_2 (waste product released from photolysis). The ATP and NADPH are used in the Calvin cycle.



Cyclic vs Non-cyclic Photophosphorylation

Feature	Non-cyclic	Cyclic

Photosystems used	PSI and PSII	PSI only
Products	ATP + reduced NADP + O_2	ATP only (no NADPH, no O_2)
Electron pathway	Linear (water → PSII → ETC → PSI → NADP)	Cyclic (PSI → ferredoxin → ETC → back to PSI)
Water split?	Yes (photolysis)	No
Purpose	Main pathway; produces both ATP and NADPH	Produces extra ATP when the Calvin cycle needs more ATP than NADPH (ATP:NADPH ratio adjustment)

Light-Independent Reactions — The Calvin Cycle (Stroma)

The Calvin cycle uses the ATP and NADPH from the light-dependent reactions to fix CO_2 into organic molecules. It does not directly require light, but it depends on the products of the light reactions, so it stops when light stops. The cycle has three stages:

1. **Carbon fixation:** CO_2 from the atmosphere combines with ribulose bisphosphate (RuBP, a 5C compound) in a reaction catalysed by the enzyme **RuBisCO** (ribulose bisphosphate carboxylase-oxygenase — the most abundant protein on Earth). This produces an unstable 6C intermediate that immediately splits into two molecules of **glycerate-3-phosphate (GP, 3C)**.
2. **Reduction:** GP is reduced to **triose phosphate (TP/G3P, 3C)** using ATP (provides energy by phosphorylation) and reduced NADP (provides hydrogen atoms for reduction). This is the step that directly uses the products of the light-dependent reactions.
3. **Regeneration of RuBP:** Most TP (5 out of every 6 molecules) is used to regenerate RuBP using ATP. This ensures the cycle can continue.
4. **Product:** 1 in 6 TP molecules (net 1 TP per 3 CO_2 fixed) exits the cycle. Two TP molecules (6C total) are combined to form one molecule of glucose. TP can also be converted into amino acids, fatty acids, glycerol, and other organic molecules needed by the plant.

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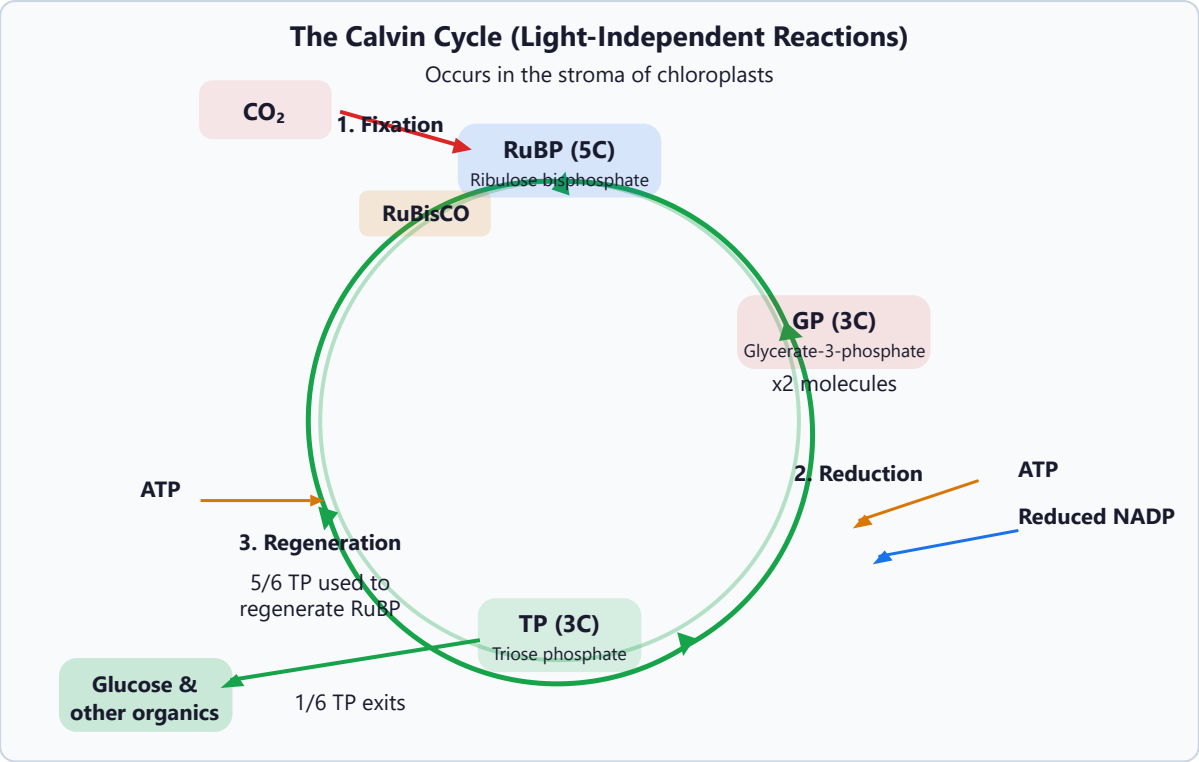

Per 3 CO_2 fixed: 3 CO_2 + 9 ATP + 6 NADPH → 1 TP (G3P) + 9 ADP + 8 P_i + 6 $NADP^+$

Per glucose: 6 CO_2 must be fixed (6 turns of the cycle, producing 2 net TP molecules that combine to form glucose)

Effect of Changing Conditions on Calvin Cycle Intermediates

Condition Change	GP Level	TP Level	RuBP Level	Explanation
Light suddenly removed	Increases	Decreases	Decreases	No ATP/NADPH produced; GP cannot be reduced to TP; RuBP not regenerated from TP; CO_2 fixation continues briefly until RuBP runs out, producing more GP
CO_2 concentration drops	Decreases	Increases (briefly)	Increases	Less CO_2 fixed with RuBP; less GP made; TP still produced from

				existing GP; RuBP regenerated but not used up (no CO_2 to react with)
Temperature increases beyond optimum	Varies	Decreases	Decreases	RuBisCO and other enzymes begin to denature; all reactions slow down

Exam Tip: Know what happens when light is suddenly removed: light-dependent reactions stop, so no ATP or reduced NADP is produced. GP accumulates (cannot be reduced to TP), TP and RuBP decrease. Conversely, if CO_2 drops: RuBP accumulates (not being used), GP decreases (less being made). These are very common 6-mark questions.

Limiting Factors

The rate of photosynthesis is limited by whichever factor is in shortest supply (Blackman's law of limiting factors):

- **Light intensity:** Rate increases proportionally at low light (directly affects the rate of the light-dependent reactions); plateaus at saturation point when another factor becomes limiting. Light intensity is inversely proportional to distance squared: $I \propto 1/d^2$
- **CO_2 concentration:** Atmospheric CO_2 (~0.04%) is often limiting in natural conditions. Increasing it raises the rate until RuBisCO is saturated (~0.1%) or another factor limits. This is why glasshouse growers enrich CO_2 levels.
- **Temperature:** Affects enzyme activity (especially RuBisCO). Optimum ~25–30°C for most temperate plants. Below optimum: insufficient kinetic energy. Above optimum: enzyme denaturation reduces rate sharply.

4.2 Cellular Respiration

Respiration: The metabolic process in which organic molecules (usually glucose) are broken down in a series of enzyme-catalysed steps to release energy, which is used to synthesise ATP from ADP and P_i. It occurs in all living cells continuously. ATP is the universal energy currency

of cells — it provides energy for active transport, muscle contraction, protein synthesis, nerve impulse transmission, and many other metabolic processes.

ATP — The Energy Currency

ATP (adenosine triphosphate) is a nucleotide derivative consisting of adenine, ribose, and three phosphate groups. Hydrolysis of the terminal phosphate bond (ATP → ADP + P_i) releases approximately 30.5 kJ mol^{-1} of free energy, which is coupled to energy-requiring reactions in the cell. ATP is rapidly recycled — a human uses approximately their own body weight in ATP per day.

Overview of Aerobic Respiration

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy (ATP + heat)}$$

Stage 1: Glycolysis (Cytoplasm)

Glycolysis is the splitting of glucose (6C) into two molecules of pyruvate (3C). It occurs in the cytoplasm of all living cells and is **anaerobic** (does not require oxygen).

1. **Phosphorylation:** Glucose (6C) is phosphorylated using 2 ATP → fructose-1,6-bisphosphate (6C). This investment of ATP makes the molecule more reactive and prevents glucose from leaving the cell (charged phosphorylated molecules cannot cross the membrane).
2. **Lysis:** Fructose-1,6-bisphosphate is split into 2 molecules of triose phosphate (TP, 3C)
3. **Oxidation:** Each TP is oxidised (hydrogen atoms removed and transferred to NAD^+, forming reduced NAD) and converted to pyruvate (3C). Substrate-level phosphorylation produces ATP directly by transferring a phosphate group from the substrate to ADP.
4. **Net yield per glucose:** 2 ATP (4 produced − 2 invested), 2 reduced NAD, 2 pyruvate

The Link Reaction (Mitochondrial Matrix)

Pyruvate enters the mitochondrial matrix via a specific transporter protein in the inner membrane. For each pyruvate molecule:

- **Oxidative decarboxylation:** pyruvate is oxidised (hydrogen removed → reduces NAD^+ to reduced NAD) and decarboxylated (one CO_2 removed — the first CO_2 released in respiration)

- The 2C acetyl group combines with **coenzyme A (CoA)** to form **acetyl CoA**

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Pyruvate (3C) + NAD+ + CoA → Acetyl CoA (2C) + CO2 + reduced NAD
(This occurs twice per glucose molecule)
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Stage 2: Krebs Cycle (Mitochondrial Matrix)

The Krebs cycle (citric acid cycle) is a series of enzyme-catalysed oxidative decarboxylation reactions:

1. Acetyl CoA (2C) combines with **oxaloacetate (4C)** to form **citrate (6C)**; CoA is released and recycled
2. Citrate is progressively decarboxylated (2 CO_2 released) and oxidised (hydrogen atoms removed and transferred to NAD^+ and FAD) through a series of intermediate compounds, regenerating oxaloacetate
3. **Per turn of the cycle:** 2 CO_2, 3 reduced NAD, 1 reduced FAD, 1 ATP (via GTP by substrate-level phosphorylation)
4. **Per glucose (2 turns):** 4 CO_2, 6 reduced NAD, 2 reduced FAD, 2 ATP

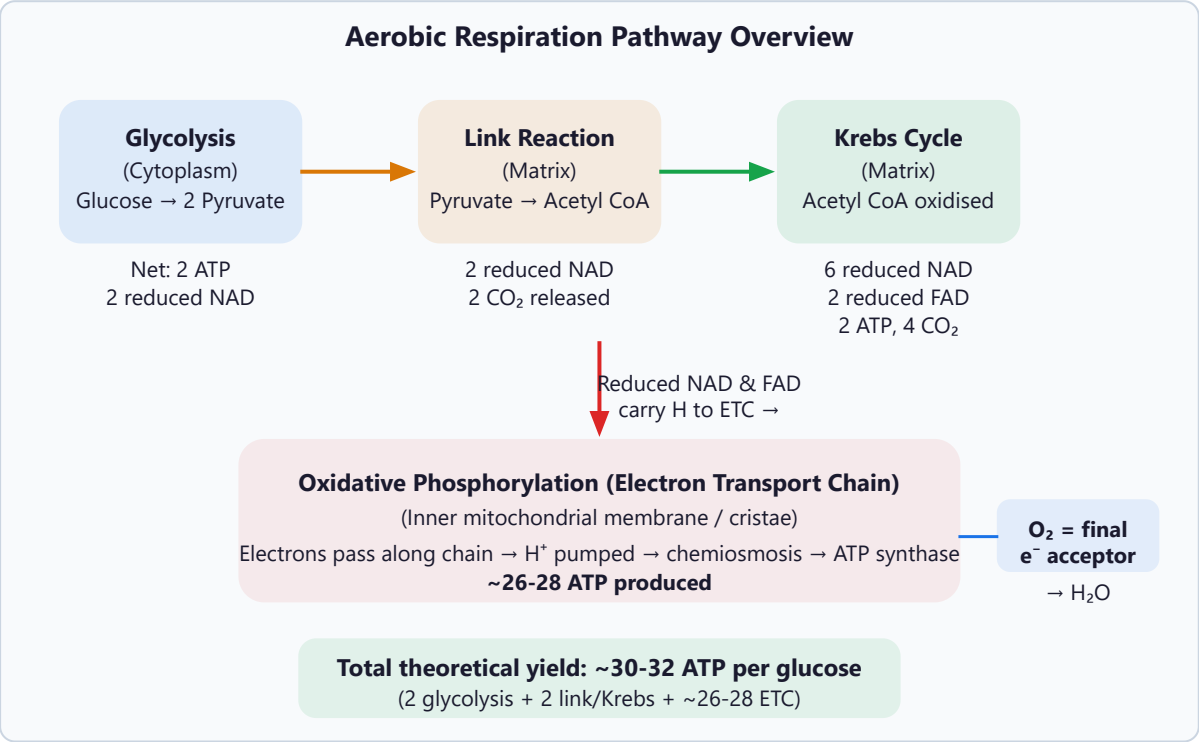
The main role of the Krebs cycle is to generate reduced coenzymes (NADH and $FADH_2$) that carry hydrogen atoms (and their electrons) to the electron transport chain for oxidative phosphorylation, where the majority of ATP is produced.

Stage 3: Oxidative Phosphorylation (Inner Mitochondrial Membrane)

This is where the vast majority of ATP is produced (~26–28 ATP per glucose). It takes place on the **cristae** (folds of the inner mitochondrial membrane, which increase the surface area for the electron transport chain and ATP synthase).

1. Reduced NAD and reduced FAD donate their electrons (and H^+ ions) to the **electron transport chain (ETC)** — a series of protein complexes (Complexes I–IV) and mobile carriers embedded in the inner membrane
2. Electrons are passed along the chain from carrier to carrier in a series of redox reactions. At each step, electrons lose energy.
3. The energy released is used to actively pump H^+ ions from the matrix into the **intermembrane space**, creating a proton gradient (electrochemical gradient). This is **chemiosmosis**.

4. H^+ ions flow back into the matrix through **ATP synthase** (down the gradient), driving the enzyme to catalyse ADP + P_i → ATP
5. **Oxygen is the final electron acceptor:** at the end of the ETC, electrons and H^+ ions combine with O_2 to form water: $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$. Without oxygen, the ETC stops because electrons have nowhere to go.



ATP Yield Summary

Stage	Location	ATP (net)	Reduced NAD	Reduced FAD	CO_2
Glycolysis	Cytoplasm	2	2	0	0
Link reaction (×2)	Matrix	0	2	0	2
Krebs cycle (×2)	Matrix	2	6	2	4
Oxidative phosphorylation	Inner membrane	~26–28	—	—	0
Total per glucose		**~30–32**	**10**	**2**	**6**

Each reduced NAD yields approximately 2.5 ATP; each reduced FAD yields approximately 1.5 ATP via the ETC. The actual yield is lower than the theoretical maximum because some of the proton gradient energy is used for other purposes (e.g., transporting pyruvate and ADP into the mitochondria), and the inner membrane is not perfectly impermeable to protons.

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Theoretical maximum ATP yield per glucose: ~30–32 ATP (actual yield is lower
due to proton leak and transport costs)
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Anaerobic Respiration

When oxygen is absent or insufficient, the ETC stops. Reduced NAD cannot be reoxidised, so NAD^+ is not regenerated. Without NAD^+, the Krebs cycle and link reaction cannot continue (they need NAD^+ as a hydrogen acceptor). Only glycolysis can continue, but only if NAD^+ is regenerated by an alternative pathway — fermentation.

Feature	Animals (lactate fermentation)	Yeast (alcoholic fermentation)
Pathway	Pyruvate → Lactate (reduced by NADH)	Pyruvate → Ethanal + CO_2 → Ethanol (ethanal reduced by NADH)
Enzyme	Lactate dehydrogenase	Pyruvate decarboxylase (removes CO_2), then alcohol dehydrogenase
NAD^+ regenerated?	Yes — allows glycolysis to continue	Yes — allows glycolysis to continue
ATP yield	2 ATP per glucose (glycolysis only)	2 ATP per glucose (glycolysis only)
Reversible?	Yes — lactate can be converted back to pyruvate in the liver (Cori cycle) when O_2 is available	No — ethanol production is irreversible; ethanol is toxic and is excreted
CO_2 produced?	No	Yes (pyruvate is decarboxylated)

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Exam Tip: If oxygen is absent, the ETC stops, reduced NAD/FAD cannot be reoxidised, Krebs cycle stops, link reaction stops. Only glycolysis continues (using anaerobic pathways to regenerate NAD^+). In animals: pyruvate → lactate. In yeast: pyruvate → ethanal + CO_2 → ethanol. Remember: the purpose of fermentation is to regenerate NAD^+, NOT to produce ATP (glycolysis produces the ATP).

Respiratory Substrates and Respiratory Quotient (RQ)

Cells can respire substrates other than glucose. The **respiratory quotient (RQ)** indicates which substrate is being respired:

$$RQ = CO_2 \text{ produced} / O_2 \text{ consumed}$$

Substrate	RQ	Explanation
Carbohydrate (glucose)	1.0	Already partially oxidised; CO_2 out = O_2 in
Lipid (fat)	~0.7	More reduced (more H atoms); needs more O_2 to oxidise fully; yields more ATP per gram
Protein	~0.9	Intermediate level of reduction; amino groups must be removed (deamination) first
Anaerobic respiration	>1.0	CO_2 produced without proportional O_2 consumption (or no O_2 consumed at all)

4.3 Ecosystems and Energy Flow

Ecosystem: A community of organisms (biotic components) interacting with each other and their physical environment (abiotic factors) within a defined area. Ecosystems are characterised

by energy flow (unidirectional, from sun through trophic levels) and nutrient cycling (recycling of matter).

Key Ecological Terms

Term	Definition
Population	A group of organisms of the same species living in the same area at the same time, able to interbreed
Community	All the populations of different species living and interacting in the same area
Habitat	The place where an organism lives
Ecological niche	The role of a species within its habitat, including its interactions with other species and the environment (no two species can occupy exactly the same niche — competitive exclusion principle)
Abiotic factors	Non-living factors: temperature, light, water, pH, wind, mineral ions
Biotic factors	Living factors: competition, predation, disease, mutualism

Energy Transfer Through Trophic Levels

Energy enters ecosystems as light and is fixed by producers (photosynthesis). Only about **1–3%** of light energy striking a leaf is converted to chemical energy in plants (most light is reflected, transmitted, or the wrong wavelength). Energy is transferred through food chains, but only approximately **10%** passes between trophic levels. Energy is lost at each level through:

- **Respiration** — heat energy lost to surroundings (the largest loss at every trophic level); this energy cannot be recycled
- **Excretion** — urine and faeces contain chemical energy that is not assimilated
- **Parts not eaten** — roots, bones, fur, shells are often not consumed
- **Parts not digested** — cellulose and other indigestible materials pass through the gut

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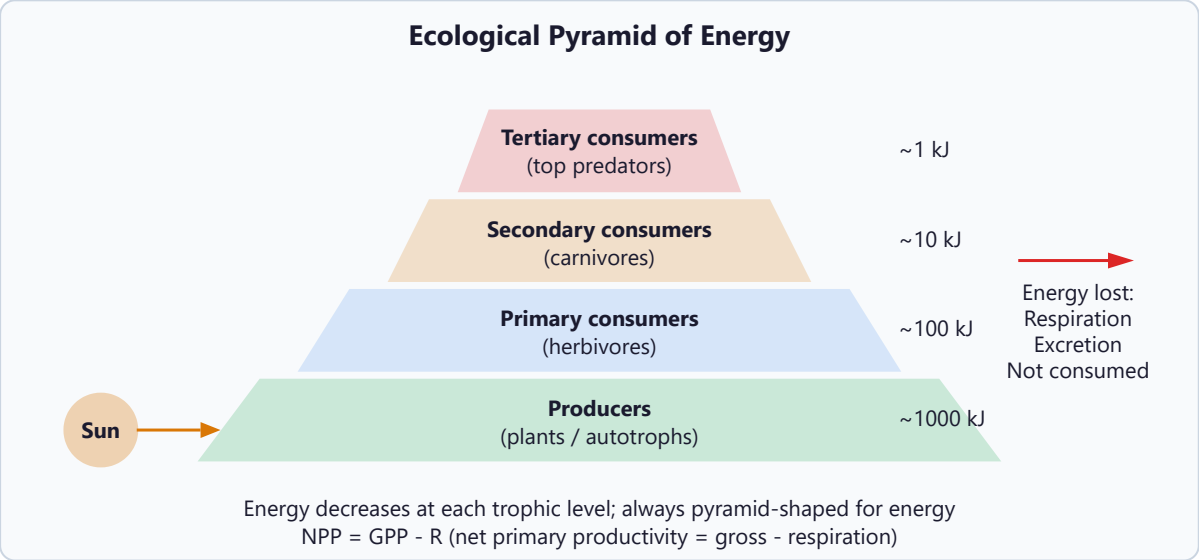
Gross primary productivity (GPP) = total rate of photosynthesis (total energy fixed by producers)

Net primary productivity (NPP) = GPP – Respiration (R) — this is the energy available to the next trophic level

Efficiency of transfer = (energy at trophic level n+1 / energy at trophic level n) × 100%



Exam Tip: Pyramids of energy are always upright (energy always decreases). Pyramids of numbers and biomass can sometimes be inverted (e.g., one oak tree supporting thousands of caterpillars; parasites being smaller than hosts). Energy cannot be recycled — it flows through the ecosystem and is ultimately lost as heat. Matter (nutrients) IS recycled.

Farming Practices and Energy Efficiency

Farmers can increase the proportion of energy transferred to humans (the desired trophic level) by:

- **Short food chains** — eating plants directly (vegetarian diet) is more energy-efficient than eating animals, as each extra trophic level loses ~90% of energy

- **Restricting animal movement** — factory farming (less energy lost through locomotion and maintaining body temperature), though this raises ethical concerns about animal welfare
- **Controlling temperature** — heated barns reduce energy lost to thermoregulation in endotherms
- **Selective breeding** — animals bred for fast growth and high food conversion ratios
- **Removing competitors** — herbicides, pesticides, and weeding reduce energy loss to organisms other than the crop

Nutrient Cycling

Unlike energy (which flows through ecosystems and is lost as heat), nutrients are **recycled** within ecosystems. Decomposers (saprobionts) play a crucial role in returning nutrients to the abiotic environment.

The Carbon Cycle

- **CO_2 removed from atmosphere** by photosynthesis (plants, algae, cyanobacteria) and dissolving in oceans
- **CO_2 returned to atmosphere** by respiration (all organisms), combustion of fossil fuels and biomass, volcanic activity, and decomposition
- **Carbon locked in long-term stores:** fossil fuels (coal, oil, gas formed from ancient organisms over millions of years), limestone/chalk ($CaCO_3$ from marine organisms), ocean sediments, peat bogs
- **Human impacts:** burning fossil fuels releases stored carbon as CO_2; deforestation reduces photosynthetic CO_2 removal and releases carbon from decaying/burning biomass. These contribute to the enhanced greenhouse effect and climate change.

The Nitrogen Cycle

Nitrogen is essential for amino acids, proteins, nucleic acids, and ATP. Atmospheric N_2 (~78% of air) is very unreactive due to the strong triple bond, so it must be "fixed" into usable forms.

Process	Organisms	Conversion	Conditions

Nitrogen fixation	*Rhizobium* (mutualistic, in legume root nodules), *Azotobacter* (free-living in soil); also lightning and the Haber process industrially	$N_2 \rightarrow NH_4^+$ (ammonium)	Anaerobic within nodules; requires nitrogenase enzyme
Nitrification	*Nitrosomonas* then *Nitrobacter* (chemoautotrophic bacteria)	$NH_4^+ \rightarrow NO_2^- \rightarrow NO_3^-$	Aerobic; these bacteria obtain energy from these oxidation reactions
Assimilation	Plants	NO_3^- absorbed by roots (active transport) and used to synthesise amino acids, then proteins and nucleic acids	Plants are the entry point for nitrogen into food chains
Ammonification	Decomposers (saprobionts — bacteria and fungi)	Organic N in dead organisms and waste (proteins, urea, nucleic acids) → NH_4^+	Extracellular digestion by secreted enzymes; deamination
Denitrification	Denitrifying bacteria (e.g., *Pseudomonas*)	$NO_3^- \rightarrow N_2$ (returned to atmosphere)	Anaerobic conditions (e.g., waterlogged, compacted soil); this is a loss of nitrogen from the soil and is undesirable for farmers

Eutrophication

Excess nitrates and phosphates (from fertiliser runoff or sewage) entering freshwater causes **eutrophication**:

1. Mineral ions stimulate rapid growth of algae on the water surface (**algal bloom**)
2. Dense algal growth blocks light to submerged plants, which die
3. Dead plants and algae are decomposed by aerobic bacteria
4. Bacterial decomposition consumes dissolved oxygen (high **biological oxygen demand, BOD**)
5. Oxygen levels fall dramatically, causing death of aerobic organisms (fish, invertebrates)
6. Further decomposition of dead animals creates a positive feedback loop; the water body becomes anaerobic

4.4 Decomposition

> **Decomposition:** The breakdown of dead organic matter by saprobionts (decomposers such as bacteria and fungi). Saprobionts secrete extracellular enzymes (e.g., proteases, lipases, cellulase) onto the dead material, digest it externally, and absorb the soluble products. Decomposition recycles nutrients (C, N, P, S) back into the soil and atmosphere, making them available for producers.

Factors affecting rate of decomposition:

- **Temperature** — affects enzyme activity of decomposers; faster in warm conditions; too hot = enzyme denaturation
- **Water availability** — needed as a solvent for enzyme function and as a medium for diffusion of nutrients; dry conditions slow decomposition
- **Oxygen availability** — aerobic decomposition is much faster than anaerobic; aerobic bacteria respire organic material more efficiently
- **pH** — affects enzyme activity; most decomposers prefer neutral to slightly acidic conditions
- **Type of material** — lignin and keratin decompose very slowly (few organisms have the enzymes to break them down); simple sugars and proteins decompose quickly
- **Surface area** — smaller pieces decompose faster (greater surface area for enzyme action)

4.5 Microbiology Techniques

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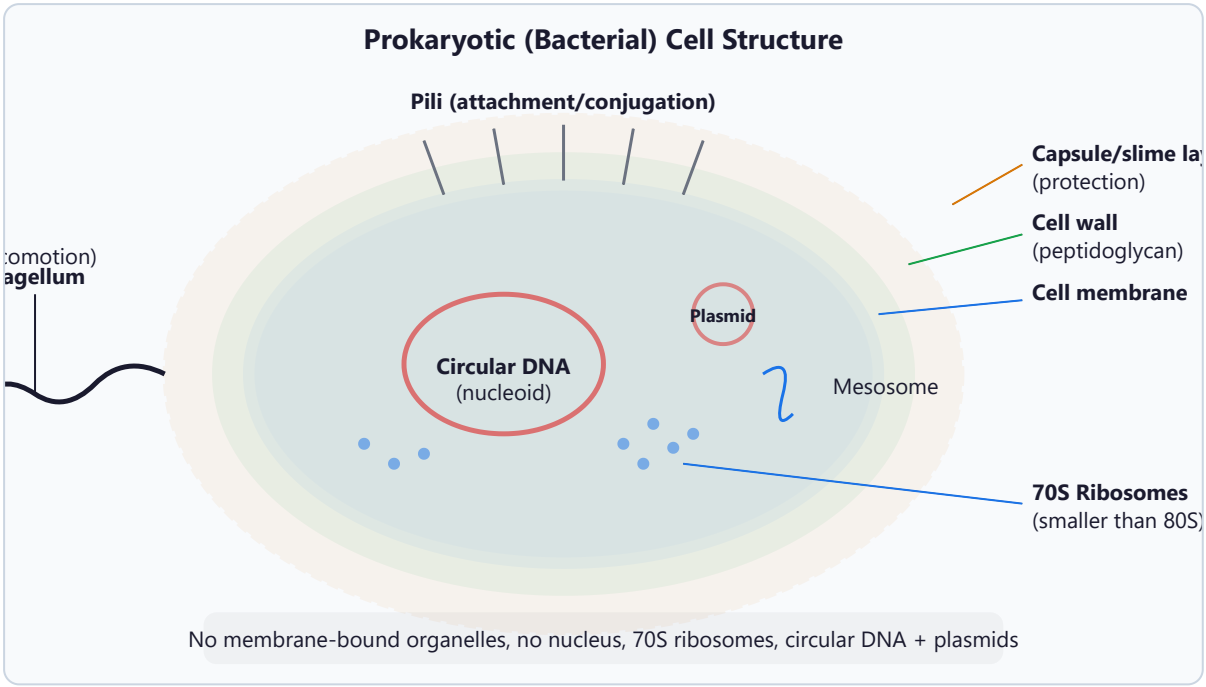
Prokaryotic Cell Structure



Comparing Prokaryotic and Eukaryotic Cells

Feature	Prokaryotic	Eukaryotic
Size	Small (0.5–5 µm)	Larger (10–100 µm)
Nucleus	No true nucleus; DNA in nucleoid region	Membrane-bound nucleus with nuclear envelope
DNA	Circular, naked (no histones); plasmids often present	Linear, associated with histone proteins; organised into chromosomes
Ribosomes	70S (smaller)	80S (larger); 70S in mitochondria and chloroplasts
Membrane-bound organelles	None	Mitochondria, ER, Golgi, lysosomes, etc.
Cell wall	Peptidoglycan (murein)	Cellulose (plants), chitin (fungi), none (animals)

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Reproduction	Binary fission (asexual)	Mitosis (asexual), meiosis (sexual)

Aseptic Technique

Essential to prevent contamination of cultures (by unwanted microorganisms) and protect the worker from potentially harmful bacteria:

- **Sterilise equipment** — autoclave at 121°C, 15 min (kills all microorganisms including endospores) or dry heat sterilisation
- **Flame inoculating loops** until red hot before and after use (kills all bacteria on the loop)
- **Work near a Bunsen burner** — the updraught of warm air reduces the chance of airborne contamination settling on cultures
- **Flame necks of bottles** when opened and before closing (kills any microorganisms on the rim and creates a convection current away from the opening)
- **Seal Petri dishes** with two small pieces of adhesive tape (not fully sealed — allows aerobic conditions to prevent growth of anaerobic pathogens like *Clostridium*)
- **Incubate at ≤25°C** in school labs — this prevents growth of human pathogens that thrive at body temperature (37°C)
- **Do not open plates** after incubation; autoclave before disposal
- **Disinfect bench** before and after work; wash hands thoroughly

Measuring Microbial Growth

Bacterial growth in a closed system (batch culture) follows a characteristic curve with four phases:

1. **Lag phase** — bacteria adapting to new environment; producing enzymes needed to metabolise available nutrients; no significant increase in cell number but cells are metabolically active and growing in size
2. **Log (exponential) phase** — rapid binary fission; cell number doubles at a constant rate; unlimited nutrients; no significant competition. Growth rate is maximal. This is when bacteria are most susceptible to antibiotics (which target actively dividing cells).
3. **Stationary phase** — growth rate equals death rate; population plateaus. Nutrients become limiting; toxic waste products (acids, alcohols) accumulate; competition intensifies. Some bacteria form endospores.

4. **Death (decline) phase** — death rate exceeds growth rate; nutrients exhausted; high concentration of toxic waste. Population declines, though some resistant cells may persist.

Counting Bacteria

Method	What It Measures	Advantages	Limitations
Viable count (serial dilution and plating)	Number of living, colony-forming cells	Only counts live bacteria; each colony = 1 original cell	Time-consuming; assumes each colony arose from one cell (may be clumps); only counts culturable bacteria
Turbidity (spectrophotometer/colorimeter)	Total cell density (living + dead)	Fast; non-destructive; can monitor in real time	Cannot distinguish living from dead cells; must calibrate against viable count
Haemocytometer	Total cell count under microscope	Direct visual count; relatively quick	Cannot distinguish live from dead without staining; small sample may not be representative

Serial Dilution and Plating

To count viable bacteria: perform serial dilutions (e.g., 10-fold: 1 cm^3 of culture into 9 cm^3 sterile water, repeat), plate known volumes on agar, incubate at appropriate temperature, count colonies on plates with 30–300 colonies. Each colony arose from one viable bacterium.

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Number of bacteria per cm³ = colonies counted × dilution factor × (1 /
volume plated in cm³)
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Exam Tip: In practical questions on microbiology, always explain WHY each aseptic step is necessary. Also, remember that turbidity (measured with a colorimeter/spectrophotometer) measures total cell count (including dead cells), while colony counting (viable count) only measures living cells. Count only plates with 30–300 colonies (fewer = unreliable; more = too crowded to count accurately, colonies may merge).

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